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Question 1.

```
class Base
```

```
{
```

```
protected :
```

```
int x;
```

```
};
```

```
class Derived : protected Base
```

```
{
```

```
};
```

int x will be inherited in Derived Class as

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Total Marks: 2

Answer

☐ As private

☐ As protected

☐ As public

☐ Can't be inherited

Question 1:

Total Marks: 2

Answer

What is the output of the following program if $a = 7$?

```
#include  
(int main()  
{  
    int a, *q, **p;  
    scanf("%d", &a);  
    q = &a;  
    p = &q;  
    printf("%d", **p);  
    return 0;  
}
```

☐ 7☐ b = 10☐ Reference of q.☐ Garbage value.

Total Marks: 2

Answer

Question 4.

Suppose you are given an array $arr[] = \{3, 2, 4, 10, 5, 1\}$.

You have to sort this array in a way such that first 3 elements will form a degenerate triangle and last 3 elements will form a generate triangle.

If a , b , and c are the lengths of the three sides of a triangle, then first sort the a , b , c in ascending order then check the condition $a + b \leq c$. If

this condition is satisfied, then the triangle is degenerate. And if this condition is not satisfied then the triangle is generated.

Which one of the following arrays will satisfy the criteria (first 3 elements will form a degenerate triangle and last 3 elements will form a generate triangle)?

☐ {3, 2, 1, 4, 10, 5}☐ {3, 2, 10, 1, 5, 4}☐ {1, 5, 10, 2, 4, 3}☐ None of the

Question 5.

Question Time Left 01 min 09 sec

Total Marks: 2

Answer

The postfix expression for the given Infix expression $(B*(C+D)) + (A/E)$ is:

☐ $B*CD++AE/$

☐ $BCD++AE/$

☐ $BCD*AE/+$

☐ $BCD*AE*+/$

Question 10.

What will be the output of the following code:

```
#include <stdio.h>
```

```
main()
```

```
{
```

```
int a[3][4] = {1,2,3,4,5,6,7,8,9,10,11,12};
```

```
int i,j,k=99;
```

```
for(i=0;i<3;i++)
```

```
for(j=0;j<4;j++)
```

```
if(a[i][j] < k)
```

```
k = a[i][j];
```

```
printf("%d", k);
```

```
}
```

10

11

12

13

14

15

16

17

Question Time Left: 03 min 05 sec

Total Marks: 2

Answer

☐ 1

☐ 6

☐ 4

☐ Compile Error

Question 11.

What will be the output of following php code?

```
<?php
```

```
echo "This "."string "."is "."separated "."by "."dots.\n";
```

```
echo "This ","string ","is ","separated ","by ","commas.";
```

```
?>
```

Total Marks: 2

Answer

☐ Error

☐ This string is separated by dots. This string is separated by commas.

☐ This string is separated by dots. This string is separated by commas.

☐ This string is separated by dots. This string is separated by commas.

Total Marks: 2

Answer

There are 'n' number of tanks arranged in a queue, filled with water. You need to collect maximum amount of water from those tanks such that no two consecutive tank is used. Also once you move forward you cannot turn back.

Which is the most efficient algorithm to solve this problem?

☐ Dynamic programming☐ Backtracking☐ Divide & Conquer☐ Stack

Question 12.

Total Marks: 2

Answer

The Big O notation is used to describe time complexity of any algorithm. What does Big O define?

☐ limiting

☐ upper

☐ ch

☐

Total Marks: 2

Answer

Question 13.

Find the time complexity of the following iterative code.

```
int i=0, j=N; // N is any positive integer
```

```
while(j>0)
```

```
{
```

```
  i=i*j;
```

```
  j=j/2;
```

```
}
```

☐ $O(1)$ ☐ $O(N)$ ☐ $O(\log N)$ ☐ $O(\sqrt{N})$

Question 14.

Total Marks: 2

Answer

You are managing an event ticketing system and need to process ticket requests in the order of their priority. Each ticket request has a priority number associated with it.

Code snippet:

```
class Ticket:
def __init__(self, priority):
self.priority = priority
```

```
tickets = []
tickets.append(Ticket(3))
tickets.append(Ticket(1))
tickets.append(Ticket(2))
tickets.append(Ticket(5))
```

```
priority_queue = []
for ticket in tickets:
heapq.heappush(priority_queue, (ticket.priority, ticket))
```

```
processed_tickets = []
while priority_queue:
processed_tickets.append(heapq.heappop(priority_queue)[1])
```

Which data structure is used to process ticket requests based on their priority in the above scenario?

☐ Stack

☐ Queue

☐ Priority Queue

☐ Linked List

Question 16.

Total Marks: 2 ⓘ

Answer

The grouchy uncle in the colony has fresh plans for this Halloween season. He has rated the children based on how they behaved over the course of the year and decided that if two kids are in line next to each other and one of them has a higher rating than the other, the kid with the higher rating would receive more candy. However, he made the decision that each youngster would receive at least one sweet because they had travelled a great distance. Find out how few candies the uncle needs to keep all of his requirements intact. The line is presented as an array with the following values: $A = [4, 1, 2, 4, 3, 1, 3]$ where $A[i]$ is the rating provided to the i th kid.

☐ 13

☐ 11

☐ 6

☐ 7

Total Marks: 2

Answer

Question 16.

You are working on a program that requires counting the number of vowels in a given string. You decide to use an array to store the vowels and compare each character of the string with the elements in the array.

```
char vowels[] = {'a', 'e', 'i', 'o', 'u'};
int count = 0;
string input = "Hello, World!";
for (int i = 0; i < input.length(); i++) {
    for (int j = 0; j < 5; j++) {
        if (input[i] == vowels[j]) {
            count++;
            break;
        }
    }
}
```

What will be the value of the variable 'count' after executing the given code snippet for the input string "Hello, World!"?

☐ 2☐ 3☐ 4☐ 5

Question 19.

Question Time Left 01:11:11

Total Marks: 2

Answer

Which row and direction arrays are proper for visiting the graph's nodes on the left, right, top, bottom, top-left, top-right, bottom-left, and bottom-right?

- ☐ `int dRow[] = {0, 0, 1, -1, -1, 1, 1, -1};`
`int dCol[] = {1, -1, 0, 0, 1, -1, -1, 1};`
- ☐ `int dRow[] = {0, -1, 1, -1, -1, 1, 1, -1};`
`int dCol[] = {1, -1, 0, 0, 1, -1, -1, 1};`
- ☐ `int dRow[] = {0, 0, 1, -1, -1, 1, 1, -1};`
`int dCol[] = {1, 1, 0, 0, 1, -1, -1, 1};`
- ☐ `int dRow[] = {0, 0, 1, 1, 1, 1, 1, 1};`
`int dCol[] = {1, 1, 0, 0, 1, 1, 1, 1};`

Question 20.

Total Marks: 2

Answer

What is the most effective algorithm for tackling this problem:

You are handed a knapsack with a weight capacity of 90 pounds. There are four components with weights of 20, 30, 40, and 70 as well as values of 70, 80, 90, and 200. What is the most valuable item you can carry in your knapsack?

☐ Greedy Algorithm☐ Kadane's Algorithm☐ Floyd's Algorithm☐ None

Bits Flipper

You are given three integers P , Q and R .

Your task is to convert these numbers into binary representation and find out the least number of bits that need to be flipped in P and Q such that the XOR operation between P and Q results in R .

Input Format

The input consists of three lines:

- The first line contains the integer P .
 - The second line contains the integer Q .
 - The third line contains the integer R .
-

Output Format

Output a single integer which represents the minimum number of bits that need to be flipped to satisfy the condition.

```
#include <iostream>
using namespace std;

int countFlips(int P, int Q, int R) {
    int x = P ^ Q;
    int y = x ^ R;
    int flips = 0;
    while (y > 0) {
        flips += y & 1;
        y >>= 1;
    }
    return flips;
}

int main() {
    int P, Q, R;
    cin >> P >> Q >> R;
    cout << countFlips(P, Q, R) << endl;
    return 0;
}
```


Question 27

Max. score: 100.00

The best subsequence

You are given the following:

- An array a of size n
- An Integer k

The special value of a subsequence p is equal to $\sum_{i=2}^k |p_i - p_{i-1}|$ where p_i denotes the i^{th} element of the subsequence p .

Find out the minimum special value among all the subsequences of array a having length k , if not possible to apply operations print 0.

Note: A subsequence is a sequence that can be derived from the given sequence by deleting zero or more elements without changing the order of the remaining elements.

Function description

Complete the `solve` function. This function takes the

```
long long solve(int n, vector<int> a, int k) {  
  
    if (k > n) return 0;  
  
    long long min_special_value = LLONG_MAX;  
    for (int i = 0; i < (1 << n); i++) {  
        vector<int> subseq;  
        for (int j = 0; j < n; j++) {  
            if (i & (1 << j)) {  
                subseq.push_back(a[j]);  
            }  
        }  
        if (subseq.size() == k) {  
            long long special_value = 0;  
            for (int m = 0; m < k; m++) {  
                special_value += (2 * subseq[m]) -  
subseq[m];  
            }  
            min_special_value =  
min(min_special_value, special_value);  
        }  
    }  
    return min_special_value == LLONG_MAX ? 0 :  
min_special_value;  
}
```



```
1 def calculate_total(counts):
2     values = [0.20,0.40,1.00,2.00,5.00,10.00]
3
4     total = sum(count* value for count, value in zip(counts, values))
5     return round(total, 1)
6
7 input_counts = input("")
8 counts=list(map(int,input_counts.split()))
9 total_amount = calculate_total(counts)
10 print(total_amount)
```

Question 1

What is the output of the following program if a = 77

```
#include  
int main()  
{  
    int a, *q, **p;  
    scanf("%d", &a);  
    q = &a;  
    p = &q;  
    printf("%d", **p);  
    return 0;  
}
```


Question 4.

Total Marks: 2

Answer

John and Jonathan have found an array $arr1[5] = \{67, 12, 90, 23, 34\}$. Both of them want to sort it.

John uses selection sort and counts that for how many times the outer loop has run and stores the count in $x1$.

☐ 1

Jonathan uses modified bubble sort and counts that for how many times the outer loop has run and stored the count in $x2$.

☐ -1

What will be the value of $(x2 - x1)$?

☐ 0

☐ 2

Question 4.

Total Marks: 2

Max has an array $\text{arr}[5] = \{1, 8, 3, 9, 5\}$. He wants to count the number of subarrays (whose length is greater than 1) in which all the elements are coprime to each other.

How many subarrays will he get?

a, b are coprime if $\text{gcd}(a, b) = 1$.

Total Marks: 2

Answer

Question 8

The following SQL is which type of join: `SELECT USER.ID, ORDER.ID, USERNAME FROM USER, ORDER WHERE USER.ID = ORDER.ID`

- ☐ Outer Join
- ☐ Cartesian Join
- ☐ Equi-Join
- ☐ Natural Join

Question 10.

While verifying digital signature by the receiver:

- ☐ Sender's private key is used
- ☐ Receiver's private key is used
- ☐ Sender's public key is used
- ☐ Receiver's public key is used

Question 12.

There are 'n' number of tanks arranged in a queue, filled with water. You need to collect maximum amount of water from those tanks such that no two consecutive tank is used. Also once you move forward you cannot turn back.

Which is the most efficient algorithm to solve this problem?

Total Marks: 2

Question Time Left 01 min 08 sec

Answer

☐ Dynamic programming

☐ Backtracking

☐ Divide & Conquer

☐ Stack